

TESD 1420 – Advanced Java Programming (4 Credits)

Course Description

This course covers advanced Java programming concepts such as generics, data structures, search trees, and advanced JavaFX design and implementation. Students will gain a sound understanding of in- depth Java programming and will use their skills to create an advanced JavaFX layout utilizing these concepts.

Course Objectives

- Explain generics and how they are used.
- Explore common data structures.
- Work with search trees and common data related to them.
- Build a useful JavaFX layout using advanced Java programming concepts.

Course Outline

- Recursion
- Generics
- Lists, Stacks, and Queues
- Sets and Maps
- Sorting
- Implementing Priority Queues
- Binary Search Trees
- Advanced JavaFX

Textbook & Reading Materials

Introduction to Java Programming and Data Structures, 13th Ed (Lifetime access) (ISBN: 9780138123352)

Assignments & Assessments

Course Introduction and Standards
Meet Your Instruction Team
Rules of the Lab
CS Code of Conduct Policy
Submitting Assignments in Canvas
Taking a Screenshot
Programming Standards
GitHub Standards
Writing Standards
Exam Standards
Orientation
Syllabus Agreement
Course References
Student Data Files
Check Exercises
Module 1 Introduction and Competencies
Chapter 18 - Reading Day
Kecursion - t-unelamental
Programming 18-9
Recursion - Real World
Chapter 18 GitHub
Module 1 Exam
Checkpoint Meeting Module 1
Student Acknowledgement Statement Module 1
Module 2 Introduction and Competencies
Chapter 19 - Reading Day
Generics - Fundamental
Generics - Real World
Programming Exercise 19-9
Chapter 19 GitHub
Module 2 Exam
Checkpoint Meeting Module 2
Student Acknowledgement Statement Module 2
Module 3 Introduction and Competencies
Chapter 20 - Reading Day
Programming Exercise 20-3
Programming Exercise 20-9
Programming Exercise 20-21
Chapter 20 GitHub
Chapter 24 - Reading Day
Programming Exercise 24-1
Programming Exercise 24-3
Programming Exercise 24-5
Chapter 24 GitHub
Module 3 Exam
Checkpoint Meeting Module 3
Student Acknowledgement Statement Module 3

Module 4 Introduction and Competencies
Chapter 21 - Reading Day
Programming Exercise 21-7
Programming Exercise 21-9
Programming Exercise 21-11
Chapter 21 GitHub
Module 4 Exam
Checkpoint Meeting Module 4
Student Acknowledgement Statement Module 4
Module 5 Introduction and Competencies
Chapter 22 - Reading Day
Programming Exercise 22-1
Programming Exercise 22-3
Chapter 22 GitHub
Chapter 23 - Reading Day
Programming Exercise 23-3
Programming Exercise 23-7
Chapter 23 GitHub
Module 5 Exam
Checkpoint Meeting Module 5
Student Acknowledgement Statement Module 5
Module 6 Introduction and Competencies
Chapter 32 - Reading Day
Programming Exercise 32-3
Programming Exercise 32-13
Chapter 32 GitHub
Chapter 33 - Reading Day
Programming Exercise 33-1
Programming Exercise 33-9
Chapter 33 GitHub
Module 6 Exam
Checkpoint Meeting Module 6
Student Acknowledgement Statement Module 6
Module 7 Introduction and Competencies
Chapter 25 - Reading Day
Programming Exercise 25-1
Programming Exercise 25-3
Programming Exercise 25-11
Chapter 25 GitHub
Module 7 Exam
Checkpoint Meeting Module 7
Student Acknowledgement Statement Module 7
Module 8 Introduction and Competencies
Chapter 31 - Reading Day
Programming Exercise 31-17
Programming Exercise 31-20
Programming Exercise 31-22

Chapter 31 GitHub
Module 8 Exam
End of Course Survey

Course Completion Checkpoint Meeting
End of Course Student Acknowledgement Statement

Subject to change. Please consult your Canvas course for the most current instructions and updates.

Classroom Hours

Mo, Tu, W, Th

8:00 AM - 11:00 AM

12:00 PM - 3:00 PM

4:00 PM - 7:00 PM

Friday

8:00 AM - 11:00 AM

12:00 PM - 3:00 PM

For a full list of course hours visit: [Course Schedule](#)

Instructor Contact Information

Jason Adams — jadams@stech.edu

Dallin Savage — dsavage@stech.edu

Nathan Merrill — nmerrill@stech.edu

Office Hours: By appointment

Email is the preferred method of communication; you will receive a response within 24 hours during regular business hours.

Canvas Information

Canvas is where course content, grades, and communication will reside for this course.

- Canvas URL: stech.instructure.com
 - For Canvas passwords or any other computer-related technical support contact Student Services.
 - For Regular Hours and Weekdays call (435) 586 - 2899
 - For After Hours and Weekends call (435) 865 - 3929 (Leave a message if no response)
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Policies

Course Grading

All assignments in this course require 100% score with unlimited submissions. All quizzes require a minimum score of 80%. High School Power School Grades: Quarter student grades will be determined by student progress percentage. Faculty will use the higher percentage of either 1) quarter progress, or 2) cumulative progress for the current training plan year.

Grade Scale: The following grading scale will be used to determine a letter grade.

A : 94 - 100%	A- : 90 - 93%	B+ : 87 - 89%	B : 83 - 86%
B- : 80 - 82%	C+ : 77 - 79%	C : 73 - 76%	C- : 70 - 72%
D+ : 67 - 69%	D : 63 - 66%	D- : 60 - 62%	F : 0 - 59%

Course Policies

You are required to keep your progress and attendance at 67% minimum. You must complete this program within 150% estimated program length. You are permitted one 15 minute break every 90 minutes. If you take more than one break in a 90 minute period or your break lasts longer than 15 minutes, your attendance will be penalized. 10 consecutive absences will lead to being withdrawn from the program. Please notify your instructors about absences as soon as possible. If absence is due to illness, please email your instructors prior to end of day.

Cell Phone/Electronics: Cell phones cannot be used during class time. You may bring your personal computers to class. You must be on topic in the lab while clocked-in.

Industry Environment: Computer science typically is very sedentary. This means you may sit at a desk for long hours. Be sure to move and get what exercise you can.

Iron County High School Students: Your progress grade will be submitted every Friday to PowerSchool. You must maintain a progress percentage of 67% or over per quarter to stay in the course.

Additional Information

InformaCast Statement: Southwest Tech uses InformaCast to ensure the safety and well-being of our students. In times of emergency, such as weather closures and delays, this app allows us to promptly deliver notifications directly to your mobile devices. To stay informed and receive real-time updates, we encourage all students to sign up for notifications. Your safety is our priority, and staying connected ensures a swift response to any unforeseen circumstances. More information and directions for signing up are available at: <https://stech.edu/emergency-notifications/>

Internet Acceptable Use Policy: The student is expected to review and follow the Southwest Technical College Internet Safety Policy at: <https://stech.edu/students/policies/>

Student Code of Conduct Policy: The student is expected to review and follow the Southwest Technical College Student Code of Conduct Policy at: <https://stech.edu/students/policies/>

Accommodations: Students with medical, psychological, learning, or other disabilities desiring accommodations or services under ADA, must contact the Student Services Office. Student Services determines eligibility for and authorizes the provision of these accommodations and services. Students must voluntarily disclose that they have a disability, request an accommodation, and provide documentation of their disability. Students with disabilities may apply for accommodations, based on an eligible disability, through the Student Services office located at 757 W. 800 S., Cedar City, UT 84720, and by phone at (435) 586-2899. No diagnostic services are currently available through Southwest Technical College.

Safety and Building Maintenance: The College has developed and follows a variety of plans to ensure the safe and effective operation of its facilities and programs. The following plans are available online: 1) Facilities Operations and Maintenance Plan; 2) Technical Infrastructure Plan; and 3) Health and Safety Plan.

Withdrawals and Refunds: Please refer to the Southwest Technical College Refund Policy at: <https://stech.edu/students/policies/>

Any high school or adult student, who declares a technical training objective is eligible for admission at Southwest Technical College (Southwest Tech). Program-specific admissions requirements may exist and will be listed on the Southwest Tech website. A high school diploma or equivalent is not required for admission but is mandatory for students seeking Title IV Federal Financial Aid.

Non-Discriminatory Policy: Southwest Technical College affirms its commitment to promote the goals of fairness and equity in all aspects of the educational enterprise, and bases its policies on the idea of global human dignity.

Southwest Tech is committed to a policy of nondiscrimination. No otherwise qualified person may be excluded from participation in or be subjected to discrimination in any course, program or activity because of race, age, color, religion, sex, pregnancy, national origin or disability. Southwest Technical College does not discriminate on the basis of sex in the education programs or activities that it operates, as required by Title IX and 34 CFR part 106. The requirement not to discriminate in education programs or activities extends to admission and employment. Inquiries about Title IX and its regulations to STECH may be referred to the Title IX Coordinator, to the Department of Education, and/or to the Office for Civil rights.

If you believe you have experienced discrimination or harassment on our campus, please contact the Title IX Coordinator, Cory Estes: cestes@stech.edu, (435) 865-3938.

For special accommodations, please contact the ADA Coordinator, Cyndie Tracy: ctracy@stech.edu, (435) 865-3944.

Southwest Technical College
757 West 800 South
Cedar City, UT 84720
info@stech.edu
(435) 586-2899